

WILLIAM R. JOHNSON

SOFTWARE ENGINEER

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CooCooCaCha
wrjcareer

EXPERIENCE

CancerIQ
2022 - present

Principal Software Engineer
Promoted from Senior Software Engineer in 2026.

Built apps and backend services for our clinical risk assessment platform, focused on cancer prevention and early detection.

- Led the team in rewriting our visual family tree (genogram) builder in React and refactoring the Rails backend to support more relationship types, twins, and family members' cancer and genetic testing data.
- Researched and developed a deterministic, recursive family tree layout algorithm that preserves paternal/maternal orientation and keeps twins adjacent, improving upon our Sugiyama-based algorithm.
- Built a custom query language using parser combinators, enabling product managers to display targeted content based on patient data.
- Standardized an Excel format for patient data with a co-founder and built a Ruby import pipeline using pre-generated UUIDs and transactions to prevent partial patient imports and resume safely after failures. Imported 500K+ records with 100+ columns.
- Led improvements to our AI-assisted development workflow by grouping apps and services in one folder and writing an AGENTS.md to give AI tools codebase context and development guidelines.

World 50
2018 - 2022

Principal Software Engineer
Promoted from Senior Software Engineer in 2021.

Full-stack engineer and manager of five engineers on our private social platform for senior executives.

- Rebuilt the legacy application with React Native, sharing most of the code across web, iOS, and Android.
- Rebuilt the API with Node.js and GraphQL, using DataLoader to avoid N+1 queries.
- Proposed an engineering process, later adopted, in which temporary, cross-functional teams planned and delivered individual epics.
- Set up databases, servers, and build pipelines using AWS, Google Cloud, and CircleCI.
- Built an activity feed that let admins filter user activity across the app. Wrote a query builder that dynamically unioned activity tables and applied filters within each table and across the combined results.

TSYS
2017 - 2018

Senior Software Engineer

Brought in for React expertise to build banking apps and modernize the team's frontend architecture.

- Rebuilt our shared React component library, removing application-specific business logic and making components more modular to simplify frontend code and improve maintainability.
- Built a dynamic React form system that generated multi-step forms from configuration and constraints supplied by the backend API.
- Introduced modern React patterns, including function components, fragments, and hooks, through a series of team presentations and mentored other developers.

Koble
2016 - 2017

Co-Founder / Lead Engineer

Co-founded a platform for people to offer their skills and discover services in their community.

- Built the React Native app and backend solo, owning software architecture and implementation.
- Set up Google Cloud infrastructure, including a Kubernetes cluster, configured CI pipelines, and distributed iOS builds through TestFlight.
- Built a service to automate Let's Encrypt certificate provisioning and renewal for TLS/SSL encryption.
- Built an audit-log service that recorded Firebase change summaries in BigQuery.

Salesloft
2015 - 2016

Full-Stack Software Engineer

Developed Salesloft's sales automation platform, from database design and migrations to frontend UI.

- Built a Docker-based QA environment to test multiple app versions simultaneously on one server cluster.
- Contributed to integrating a third-party sentiment analysis service into the application's email features.
- Learned Ruby, Rails, and Angular on the job with no prior experience, contributing code on day one.

EDUCATION

Georgia Institute of Technology
B.S. in Computer Science
2007 - 2012

SKILLS

Languages: TypeScript, JavaScript, Ruby, Rust, Python
Backend & Data: Rails, Node.js, GraphQL, SQL, PostgreSQL
Frontend & Mobile: React, React Native, HTML, CSS
Cloud & Tooling: AWS, Google Cloud, Kubernetes, Git